

Topic and subtopic of AI

1. Introduction
2. Agents and the types of agents
 - a. Simple Reflex agent
 - b. Reflex agent with state
 - c. Goal based agents
 - d. Utility Based agents
 - e. Learning Based agents
3. Environment and its types
 - a. Fully observable vs. partially observable
 - b. Deterministic vs. stochastic
 - c. Episodic vs. sequential
 - d. Static vs. dynamic
 - e. Discrete vs. continuous
 - f. Single agent vs. multi-agent
4. State space and problem formulation
 - a. Problem solving Agents
 - b. Tree Search
 - c. Graph Search
5. Uninformed searches
 - a. BFS
 - b. DFS
 - c. Depth-limited Search
 - d. Iterative Deepening Search
 - e. Uniform-cost Search
 - f. Bidirectional Search)
6. Informed searches
 - a. Best First Search
 - b. A* Search
 - c. Recursively Best First Search
7. Heuristic admissibility and consistency
8. Local Searches

- a. Beyond Classical Search
 - b. Hill Climb, Simulated Annealing ,Local Beam Search
- 9. Adversarial Search - I
 - a. Mimimax
- 10. Adversarial Search - II
 - a. Alpha Beta Pruning
- 11. Constraint Specification Problem (CSP)
 - a. Formulation of CSP
 - b. Inference in CSP
 - c. Constraint in CSP
 - d. Heuristic in CSP
- 12. Knowledge Based Reasoning, Propositional logic – I
 - a. Syntax and semantic
 - b. Entailment
 - c. Inference
 - d. Resolution Theorem
 - e. CNF
 - f.
- 13. First Order Logic
 - a. Predicates
 - b. Quantifiers
 - c. Instantiation
 - d. Unification
 - e. Resolution Graph
 - f. Green Trick
- 14. Learning and its types
 - a. Supervised Learning
 - b. Unsupervised Learning
 - c. Semi- Supervised Learning
 - d. Reinforcement Learning
- 15. Unsupervised Learning
 - a. K means Clustering

- b. K medoid Clustering
- 16. ANN
 - a. Perceptron
 - b. Neural Network Components
 - c. Perceptron Training Rule
 - i. Delta Rule
 - ii. Gradient Descent
 - iii. Stochastic Gradient Descent
 - d. Multi Layer Perceptron Architecture
 - i. Sigmoid Function
 - e. Back Propagation Algorithm